Sentiment Analysis of IMDb Movie Reviews Using Logistic Regression, DistilBERT and RoBERTa

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Abstract

Sentiment analysis is an important Natural Language Processing (NLP) task used to identify opinions and emotions expressed in textual data. In this project, sentiment analysis was performed on the IMDb movie reviews dataset using both traditional machine learning and transformer-based approaches. A baseline model was implemented using TF-IDF feature extraction and Logistic Regression for binary sentiment classification. In addition, transformer-based models including DistilBERT and RoBERTa were fine-tuned on the IMDb dataset to improve contextual understanding and classification performance.

The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Attention map visualization was also implemented using DistilBERT to improve model interpretability and understand which words influenced predictions. Experimental results showed that transformer-based models captured contextual information more effectively, with DistilBERT achieving the highest overall accuracy. This project demonstrates the effectiveness of transformer architectures for sentiment analysis and highlights practical challenges such as GPU memory limitations and computational requirements during training.

Keywords

Sentiment Analysis, Natural Language Processing, IMDb Reviews, Logistic Regression, DistilBERT, RoBERTa, Transformers

1 Introduction

Sentiment analysis is a Natural Language Processing (NLP) task whose purpose is to identify and classify opinions, emotions, or attitudes expressed in textual data. With the accelerated growth of online platforms, users often share their opinions through reviews, comments, and social media posts. Analyzing this large volume of textual information manually is difficult and time-consuming, making automated sentiment analysis an important research area in machine learning and artificial intelligence.

Sentiment analysis has various real-world applications, such as product review analysis, customer feedback monitoring, recommendation systems, and market research. By automatically identifying whether a piece of text expresses a positive or negative sentiment, organizations can gain valuable insights into customer opinions and improve decision-making processes.

Traditional machine learning techniques such as Logistic Regression have been widely used for sentiment classification tasks. These approaches are typically based on feature extraction methods, such as Term Frequency–Inverse Document Frequency (TF-IDF), to convert textual data into numerical representations that can be processed by machine learning algorithms[1]. Even though these methods are computationally efficient and often achieve strong performance, they may struggle to capture the contextual relationships between words in a sentence.

Recent developments in transformer models have considerably improved the performance of many NLP tasks. Transformer models use self-attention mechanisms to capture semantic dependencies between words within a sentence[2]. Models such as DistilBERT and RoBERTa are pre-trained on large text datasets and can learn complex linguistic patterns and contextual information. Through a process known as fine-tuning, these models can be adapted to specific tasks such as sentiment analysis while utilizing the knowledge gained during pre-training.

The goal of this project is to perform sentiment analysis on the IMDb Movie Reviews dataset and compare the performance of a traditional machine learning approach with transformer-based models. Specifically, Logistic Regression with TF-IDF features is used as a baseline model, while DistilBERT and RoBERTa are fine-tuned for binary sentiment classification. The models are evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score, to determine their effectiveness in predicting review sentiment.

2 Related Work

Sentiment analysis has been commonly studied in the fields of Natural Language Processing (NLP) and machine learning. The early approaches mainly relied on traditional machine learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression combined with feature extraction techniques including Bag-of-Words and TF-IDF [3, 4]. These methods demonstrated strong performance on sentiment classification tasks while remaining computationally efficient.

Recent innovations in deep learning have led to the development of transformer-based architectures that significantly improve the understanding of semantic information in text. Models such as BERT, DistilBERT, and RoBERTa are pre-trained on large-scale text datasets and can be fine-tuned for downstream NLP tasks, including sentiment analysis [5–7]. These models have achieved state-of-the-art results on numerous benchmark datasets by effectively capturing semantic and linguistic patterns in words.

Various studies have reported that transformer-based models generally achieved higher precision compared to traditional machine learning approaches on sentiment classification tasks [5, 7]. Motivated by these findings, this study compares a traditional baseline model, Logistic Regression with TF-IDF features, against two transformer-based models, DistilBERT and RoBERTa, on the IMDb Movie Reviews dataset.

3 Methodology

3.1 Dataset

Table 1 summarizes the main statistics of the IMDb Movie Reviews dataset used in this study.

Item	Value
Total Reviews	50,000
Positive Reviews	25,000
Negative Reviews	25,000
Training Samples	40,000
Validation Samples	5,000
Testing Samples	5,000
Task Type	Binary Classification

Table 1: Dataset Statistics

The experiments performed in this study utilize the IMDb Movie Reviews dataset, a widely used standard dataset for sentiment analysis tasks. The dataset contains a total of 50,000 movie reviews collected from the Internet Movie Database (IMDb). Each review is manually labeled as either positive or negative, making it suitable for binary sentiment classification.

The dataset is balanced, containing 25,000 positive reviews and 25,000 negative reviews. This balanced distribution helps reduce potential bias during model training and ensures a fair evaluation of classification performance. The reviews vary considerably in length and writing style, providing a realistic representation of user-generated content encountered in real-world applications.

To optimize model development and evaluation, the dataset was divided into training, validation, and testing subsets using a stratified sampling strategy. Stratified splitting was employed to preserve the original class distribution across all subsets. The final split consisted of 40,000 reviews for training, 5,000 reviews for validation, and 5,000 reviews for testing.

The IMDb dataset was selected because of its popularity in sentiment analysis research, its balanced class distribution, and its suitability for evaluating both traditional machine learning methods and transformer-based deep learning models.

3.2 Data Preprocessing

Before training the machine learning and transformer-based models, several preprocessing steps were applied to improve the quality and consistency of the textual data. Raw movie reviews often contain unnecessary symbols, formatting tags, and inconsistencies that can negatively affect model performance.

The first preprocessing step involved converting all text to lowercase in order to maintain consistency across reviews and reduce vocabulary size. HTML tags present in some reviews were removed because they do not provide meaningful information to sentiment classification. Special characters, punctuation marks, and unnecessary whitespace were also removed to reduce noise in the dataset.

After text cleaning, sentiment labels were encoded into numerical values, where positive reviews were represented by 1 and negative reviews by 0. This numerical representation is required for machine learning and deep learning models.

The preprocessed dataset was then used for feature extraction in the Logistic Regression model and tokenization in the transformer-based models.

3.3 Exploratory Data Analysis

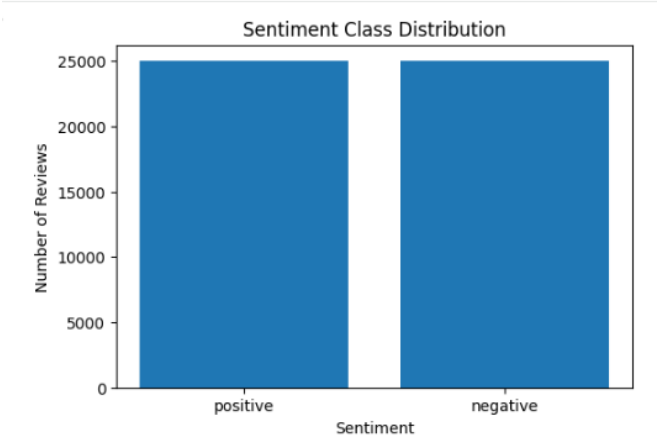


Figure 1: Class distribution of positive and negative reviews in the IMDB dataset

Exploratory Data Analysis (EDA) was carried out to better understand the characteristics of the IMDb Movie Reviews dataset before model training. One of the main goals of EDA was to analyze the distribution of sentiment classes within the dataset.

Figure 1 illustrates the class distribution of positive and negative reviews. The dataset contains an equal number of positive and negative samples, with 25,000 reviews belonging to each class. This balanced distribution is beneficial for sentiment classification tasks because it reduces the likelihood of model bias toward a particular class and enables more reliable performance evaluation.

The even distribution of the dataset ensures that classification metrics such as accuracy, precision, recall, and F1-score provide meaningful insights into model performance. Based on this analysis, no additional class balancing techniques were required before model training.

In addition to class distribution analysis, review length characteristics were examined during exploratory data analysis. The IMDb dataset contains reviews of varying lengths, ranging from short comments to detailed and lengthy movie evaluations. This variation introduces additional complexity to the sentiment classification task because models must learn to identify sentiment patterns across both concise and extensive textual inputs.

Understanding review length variation is important when working with transformer-based models, as sequence length directly influences tokenization, padding, truncation, and computational requirements during training. The observed diversity in review lengths motivated the use of fixed sequence lengths and appropriate preprocessing techniques to ensure efficient model training and evaluation.

3.4 Baseline Model: TF-IDF and Logistic Regression

Logistic Regression was used as the baseline machine learning model for sentiment classification. As machine learning algorithms cannot directly process textual data, feature extraction was performed using the Term Frequency–Inverse Document Frequency (TF-IDF) technique.

TF-IDF is a widely used feature extraction technique that measures the importance of words within documents [1]. TF-IDF converts movie reviews into numerical feature vectors by assigning importance scores to words based on their frequency within a review and across the dataset. To reduce computational complexity, the top 5,000 TF-IDF features were selected for model training.

The TF-IDF score is calculated as:

$$TF\text{-}IDF(t,d) = TF(t,d) \times IDF(t)$$

where $TF(t,d)$ represents the frequency of term t in document d, and $IDF(t)$ is calculated as:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Here, N represents the total number of documents in the dataset, and $df(t)$ represents the number of documents containing term t.

After TF-IDF vectorization, the resulting feature vectors were used to train a Logistic Regression classifier. The model was trained to classify each review as either positive or negative. This baseline model provides a reference point for comparing the performance of transformer-based models such as DistilBERT and RoBERTa.

3.5 Transformer-based models

Transformer-based models are advanced deep learning architectures widely used in Natural Language Processing (NLP) tasks. Unlike traditional machine learning approaches, transformer models use self-attention mechanisms to capture contextual relationships between words in a sentence. In this project, DistilBERT and RoBERTa were used for sentiment classification due to their strong contextual understanding and high performance in text analysis tasks.

3.5.1 ***Transformer Architecture and Self-Attention.*** Transformer architectures have become highly important in modern Natural Language Processing tasks because of their ability to process contextual information efficiently [2]. Unlike traditional sequential models such as Recurrent Neural Networks (RNNs), transformers process all words in a sentence simultaneously using a mechanism called self-attention.

Self-attention allows the model to assign different importance weights to words depending on their contextual relevance within a sentence[2]. This mechanism enables transformer models to better understand long-range dependencies and semantic relationships between words. For example, in sentiment analysis, the sentiment of a sentence often depends on combinations of words rather than isolated keywords[8].

Transformer models also utilize multi-head attention, which enables the model to focus on different linguistic relationships at the same time. In addition, positional encoding is used to preserve information about word order within the sentence because transformers process input tokens in parallel[2].

The strong contextual understanding provided by transformer architectures has led to significant improvements in NLP tasks such as sentiment analysis, machine translation, question answering, and text summarization. In this project, DistilBERT and RoBERTa were selected because they are transformer-based models capable of capturing complex semantic and contextual information from movie reviews.

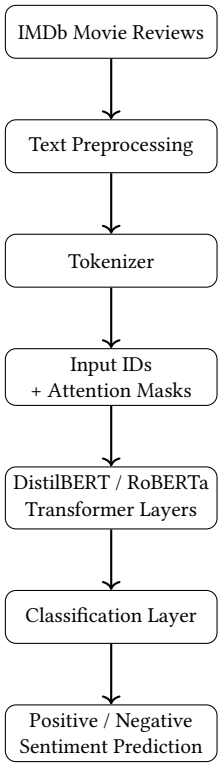


Figure 2: Pipeline of the sentiment classification system. Raw IMDb reviews are preprocessed, tokenized, converted into input representations, processed by transformer layers, and classified into positive or negative sentiment.

3.5.2 ***DistilBERT.*** DistilBERT is a lightweight and computationally efficient version of BERT designed for Natural Language Processing tasks. It is based on transformer architecture and utilizes self-attention mechanisms to capture semantic relationships between words within textual data. DistilBERT was developed using a knowledge distillation technique, where a smaller student model learns from a larger pretrained BERT model while retaining most of its language understanding capabilities.

Compared to the original BERT architecture, DistilBERT contains fewer parameters, making it faster and more memory-efficient while still achieving strong performance across various NLP tasks.

In this project, DistilBERT was fine-tuned on the IMDb movie reviews dataset for binary sentiment classification. The pretrained

tokenizer converted movie reviews into tokenized input sequences, which were then processed by the transformer model during training and evaluation. Experimental results demonstrated that DistilBERT achieved the highest classification accuracy among all implemented models while maintaining lower computational requirements compared to RoBERTa.

The strong performance of DistilBERT highlights the effectiveness of transformer-based architectures in capturing semantic and contextual information from textual data. Despite being a distilled version of BERT, the model successfully identified sentiment-related patterns within movie reviews and produced highly accurate sentiment predictions[8].

3.5.3 ***RoBERTa.*** RoBERTa is an optimized transformer-based language model developed as an improved version of BERT. The model utilizes transformer architecture and self-attention mechanisms to learn contextual and semantic relationships between words within textual sequences. RoBERTa improves upon the original BERT training strategy through modifications such as larger training datasets, dynamic masking, and longer training duration.

Because of these optimizations, RoBERTa has achieved strong performance across many Natural Language Processing tasks, including sentiment analysis, question answering, and text classification. However, the model contains a larger architecture and requires greater computational resources compared to lightweight transformer models such as DistilBERT.

In this project, RoBERTa was fine-tuned on the IMDb movie reviews dataset for binary sentiment classification. The pretrained tokenizer was used to convert movie reviews into tokenized sequences, which were processed by the transformer model during training. Experimental results showed that RoBERTa achieved strong sentiment prediction capability and demonstrated effective contextual understanding of positive and negative movie reviews.

Although RoBERTa achieved high classification accuracy, training the model required significantly more GPU memory and longer execution times compared to DistilBERT. Runtime interruptions and memory allocation issues were occasionally encountered during training in Google Colab due to hardware limitations. These observations highlight the trade-off between model complexity and computational efficiency when working with large transformer architectures.[8].

3.5.4 ***Tokenization and Input Representation.*** Before training the transformer-based models, the textual reviews were converted into tokenized input sequences using pretrained tokenizers provided by the Hugging Face Transformers library. Tokenization is the process of splitting text into smaller units called tokens, which are then converted into numerical input IDs that can be processed by transformer models.

Because movie reviews vary considerably in length, padding and truncation techniques were applied to ensure consistent input sizes within training batches. Padding adds special tokens to shorter sequences, while truncation limits very long reviews to a fixed maximum sequence length. In addition, attention masks were generated to help the transformer models distinguish between actual text tokens and padded tokens during training.

These preprocessing steps are essential for transformer architectures because they enable efficient batch processing and proper handling of variable-length textual data. Figure 2 illustrates the overall workflow of the sentiment analysis system, from raw IMDb reviews to the final sentiment prediction.

4 Experiments and Evaluation

4.1 Experimental Setup

Table 2: Hyperparameter Settings

Parameter	Value
Training Samples	40,000
Validation Samples	5,000
Testing Samples	5,000
Epochs	2
Learning Rate	2×10^{-5}
Batch Size	4 / 8
Maximum Sequence Length	128 / 256
Optimizer	AdamW

The experiments were conducted using Google Colab with access to an NVIDIA T4 GPU for accelerating model training and evaluation. The implementation was developed using Python and several machine learning libraries, including Scikit-learn, PyTorch, Hugging Face Transformers, NumPy, and Pandas.

The IMDb dataset was divided into training, validation, and testing subsets containing 40,000, 5,000, and 5,000 reviews, respectively. A stratified sampling strategy was applied to preserve the original class distribution across all subsets.

For the Logistic Regression model, TF-IDF feature extraction was applied using the top 5,000 features. Transformer-based models were fine-tuned using the AdamW optimizer and pretrained tokenizers provided by the Hugging Face library. The hyperparameter settings used throughout the experiments are presented in Table 2.

Because of computational limitations and training time constraints, extensive hyperparameter tuning was not performed. Instead, commonly recommended settings from prior literature and Hugging Face documentation were adopted.

4.2 Evaluation Metrics

To evaluate the performance of the sentiment classification models, four standard classification metrics were used: accuracy, precision, recall, and F1-score. These metrics provide a detailed analysis of model effectiveness for binary classification tasks and help compare the performance of traditional machine learning and transformer-based approaches.

Accuracy measures the proportion of correctly classified reviews among all predictions made by the model. It provides an overall indication of model effectiveness and is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

Precision evaluates the proportion of reviews predicted as positive that are actually positive. High precision indicates that the model produces fewer false positive predictions. Precision is calculated as:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of actual positive reviews that are correctly identified by the model. A high recall value indicates that the model successfully detects most positive samples. Recall is calculated as:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall into a single metric by calculating their harmonic mean. This metric is especially useful when evaluating predictive performance comprehensively because it balances both false positives and false negatives. The F1-score is calculated as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

In addition to these metrics, confusion matrices were generated to provide a detailed visualization of classification performance. Confusion matrices help identify the number of correctly and incorrectly classified reviews for each sentiment class and provide further insight into the types of prediction errors made by the models.

4.3 Training Challenges

During model training, several practical implementation challenges were encountered, particularly while fine-tuning transformer-based architectures. Large models such as RoBERTa required substantial GPU memory and computational resources, which occasionally resulted in runtime crashes and memory allocation issues in Google Colab.

To address these limitations, smaller batch sizes and mixed precision training techniques were utilized to improve training stability and reduce memory usage. In addition, training transformer models required significantly longer execution times compared to traditional machine learning approaches such as Logistic Regression.

5 Results and Discussion

5.1 Logistic Regression Results

The Logistic Regression model achieved an accuracy of 89.28% on the test set.

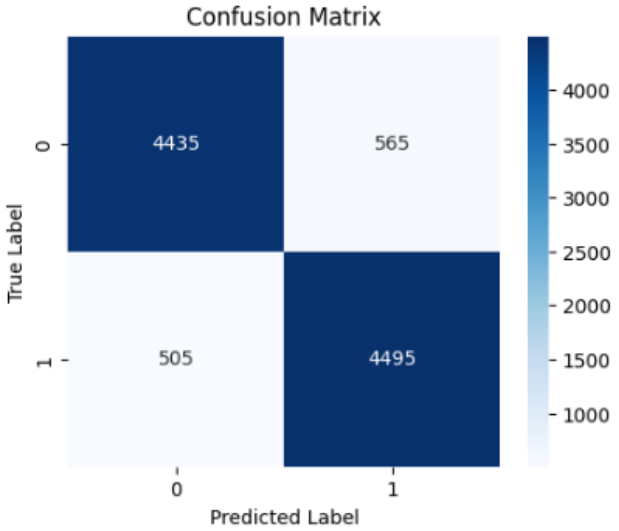


Figure 3: Confusion matrix for Logistic Regression

As shown in Figure 3, the model correctly classified most positive and negative reviews. Further implementation details are provided in Section 3.4. Although Logistic Regression is considered a relatively simple machine learning algorithm compared to transformer-based architectures, it still achieved strong performance on the IMDb dataset. This result demonstrates that TF-IDF feature extraction can effectively capture important sentiment-related words and phrases within textual data. However, the model mainly relies on word frequency information and may struggle to fully understand contextual meaning and semantic relationships between words in more complex reviews.

5.2 DistilBERT Results

The fine-tuned DistilBERT model achieved strong performance on the IMDb movie reviews dataset with an accuracy of approximately 92.30%. The model also showed balanced precision, recall, and F1-score values, indicating effective classification of both positive and negative reviews. The confusion matrix demonstrated that most reviews were correctly classified, with relatively few misclassifications. These results highlight the effectiveness of transformer-based architectures for sentiment analysis tasks.

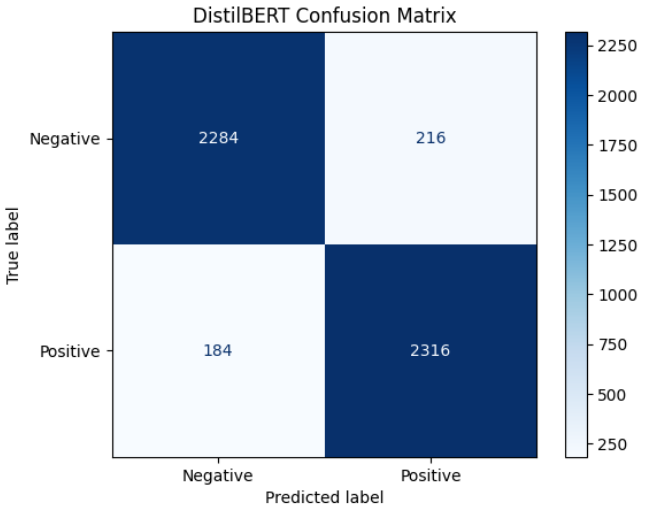


Figure 4: Confusion matrix of the DistilBERT model on the IMDb movie reviews dataset. The matrix shows the number of correctly and incorrectly classified positive and negative reviews, demonstrating the strong classification performance of the transformer-based model.

5.3 RoBERTa Results

The fine-tuned RoBERTa model achieved an accuracy of approximately 91.28% on the IMDb movie reviews dataset. The model demonstrated effective sentiment classification performance for both positive and negative reviews. However, RoBERTa required higher computational resources and longer training time compared to DistilBERT. Despite these challenges, the model produced strong overall results and further demonstrated the effectiveness of transformer-based architectures for sentiment analysis tasks. Compared to DistilBERT, RoBERTa required significantly more computational resources during training because of its larger architecture and higher model complexity. The increased GPU memory consumption occasionally resulted in runtime interruptions and slower training speed in Google Colab. These practical implementation challenges highlight one of the major limitations of large transformer models when computational resources are limited.

5.4 Model Comparison

Model	Accuracy
Logistic Regression + TF-IDF	89.28%
RoBERTa	91.28%
DistilBERT	92.30%

Table 3: Model Performance Comparison

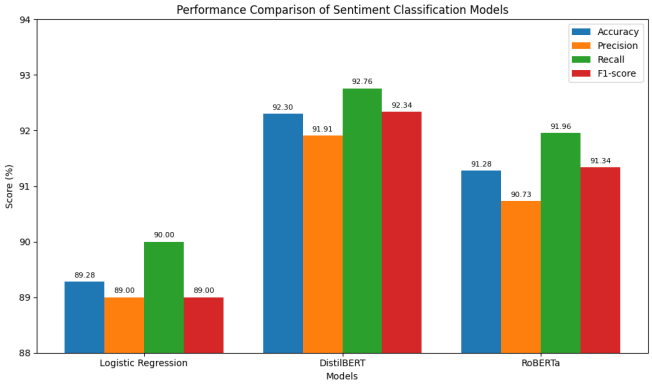


Figure 5: Comparison of accuracy, precision, recall, and F1-score for Logistic Regression, DistilBERT, and RoBERTa on the IMDb sentiment classification dataset.

The experimental results demonstrate that transformer-based models achieved stronger sentiment classification performance compared to the traditional machine learning baseline. This improvement is mainly due to the ability of transformer architectures to capture contextual and semantic relationships between words through self-attention mechanisms.

DistilBERT achieved the highest accuracy while also requiring lower computational resources compared to RoBERTa. Although RoBERTa demonstrated effective semantic understanding, its larger architecture resulted in increased GPU memory usage and longer training times. Logistic Regression with TF-IDF performed reasonably well; however, it relied primarily on word frequency information and could not fully capture contextual meaning within complex reviews.

5.5 Attention Mapping and Explainability

Transformer models use self-attention mechanisms to understand contextual relationships between words in a sentence. In this project, attention map visualization was implemented using the fine-tuned DistilBERT model to improve interpretability and understand which words influenced the sentiment prediction.

Attention maps visualize the attention weights assigned to different tokens during prediction. Darker regions in the attention map indicate higher attention values, showing which words the model focused on more strongly while classifying the review as positive or negative. This explainability technique helps provide insight into the model's decision-making process and demonstrates how transformer models capture important sentiment-related information from textual data.

Self-attention mechanisms allow transformer models to assign different importance weights to words within a sentence depending on their contextual relevance. Unlike traditional sequential models, transformers can process relationships between words simultaneously, enabling better contextual understanding of long textual sequences. This capability is particularly beneficial for sentiment analysis because the sentiment of a sentence often depends on the relationship between multiple words rather than individual keywords

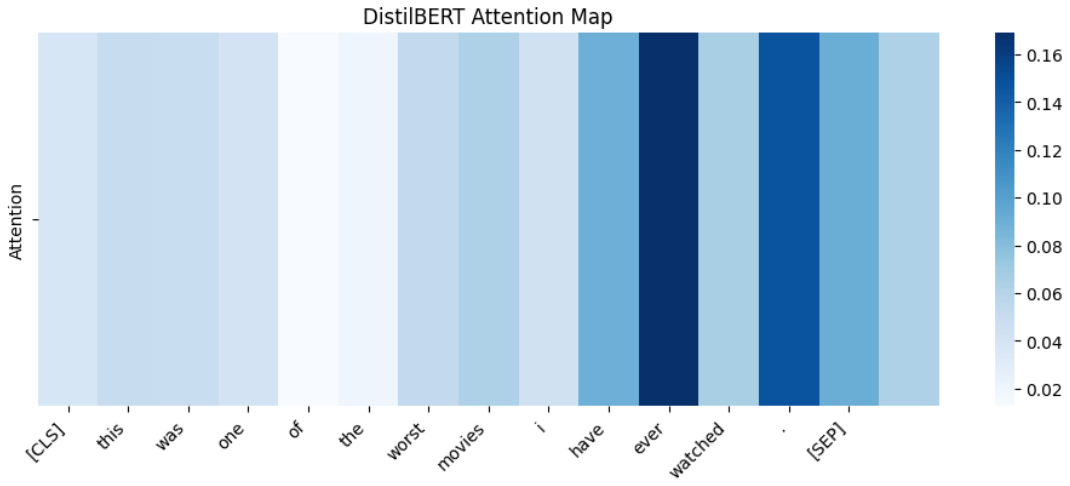


Figure 6: DistilBERT attention map visualization highlighting the words that contributed most strongly to sentiment prediction.

5.6 Discussion

The experimental results demonstrate clear performance differences between traditional machine learning approaches and transformer-based architectures. Logistic Regression combined with TF-IDF achieved strong baseline performance while remaining computationally efficient. However, the model relied primarily on statistical word representations and was less effective at capturing contextual meaning within complex reviews.

Transformer-based models achieved stronger classification performance because of their ability to model semantic and contextual relationships using self-attention mechanisms. DistilBERT achieved the best balance between classification accuracy and computational efficiency, while RoBERTa demonstrated strong contextual understanding but required higher computational resources and longer training times.

The attention map visualizations further highlighted the interpretability advantages of transformer architectures by identifying the words that contributed most strongly to sentiment predictions. These observations demonstrate the effectiveness of transformer-based NLP approaches for sentiment analysis tasks involving large textual datasets.

The comparison between traditional machine learning and transformer-based architectures also highlights the evolution of sentiment analysis techniques in Natural Language Processing. Traditional approaches such as TF-IDF and Logistic Regression primarily depend on statistical word representations, which may not fully capture semantic dependencies within complex textual data. In contrast, transformer architectures utilize self-attention mechanisms to analyze relationships between words within an entire sentence, enabling stronger contextual and semantic understanding.

Another important observation from the experimental results is the balance between computational efficiency and classification performance. Although RoBERTa demonstrated strong contextual modeling capabilities, DistilBERT achieved comparable or better performance while requiring fewer computational resources and shorter training times. This suggests that lightweight transformer architectures can provide practical advantages in real-world applications where hardware resources and inference speed are important considerations.

5.7 Practical Implications

The findings of this project demonstrate that transformer-based NLP models can be effectively applied to real-world sentiment analysis tasks involving large amounts of textual data. Automated sentiment classification systems can support applications such as customer feedback analysis, product review monitoring, recommendation systems, and social media opinion mining.

The results also highlight the trade-off between computational efficiency and model performance. While transformer-based models achieved higher classification accuracy, they required greater computational resources and longer training times compared to traditional machine learning approaches. In practical environments with limited hardware resources, lightweight transformer models such as DistilBERT may provide a better balance between performance and efficiency.

In addition, attention map visualization improves the interpretability of transformer models by providing insight into how predictions are generated. Explainability is becoming increasingly important in modern AI systems because it helps improve transparency, reliability, and user trust in automated decision-making systems.

6 Limitations

Despite achieving strong performance, this project has several limitations. Transformer-based models such as DistilBERT and RoBERTa require significant computational resources, GPU memory, and longer training times compared to traditional machine learning models. During implementation, training large transformer models caused runtime crashes and memory limitations in Google Colab, especially for RoBERTa. Due to these hardware constraints,

extensive hyperparameter tuning and experimentation with larger architectures were limited.

Another limitation is that the project focused only on binary sentiment classification using English movie reviews from the IMDb dataset. As a result, the models were not evaluated on multilingual data or more complex sentiment analysis tasks such as aspect-based sentiment classification. In addition, although attention maps improved interpretability, they do not provide a complete explanation of transformer model decisions. Finally, the performance of transformer models may vary when applied to different domains or datasets outside movie reviews.

7 Conclusion

In this project, sentiment analysis was successfully performed on the IMDb Movie Reviews dataset using both traditional machine learning and transformer-based approaches. Logistic Regression combined with TF-IDF feature extraction was implemented as a baseline model, while DistilBERT and RoBERTa were fine-tuned as transformer-based architectures for binary sentiment classification.

Experimental results demonstrated that transformer-based models achieved higher classification performance compared to the traditional machine learning baseline. DistilBERT achieved the highest overall accuracy while also maintaining lower computational requirements compared to RoBERTa. The results indicate that transformer architectures are highly effective at capturing contextual and semantic relationships within textual data, making them well suited for sentiment analysis tasks.

The project also highlighted the importance of self-attention mechanisms in improving semantic understanding of movie reviews. In addition, attention map visualization provided insight into the decision-making behavior of transformer models by identifying which words influenced sentiment predictions. This explainability component improved the interpretability of the transformer-based models and demonstrated how attention mechanisms contribute to classification performance.

From a practical perspective, the project provided valuable experience in Natural Language Processing workflows, including data preprocessing, TF-IDF feature extraction, transformer tokenization, fine-tuning pretrained language models, model evaluation, and GPU optimization. Several implementation challenges were encountered during transformer training, particularly related to GPU memory limitations and runtime constraints in Google Colab. These challenges further emphasized the computational demands of modern transformer architectures.

Overall, this project demonstrates the effectiveness of transformer-based NLP models for sentiment analysis and provides a comparative analysis between traditional machine learning techniques and modern deep learning approaches. The findings suggest that transformer architectures such as DistilBERT and RoBERTa can significantly improve sentiment classification performance while also offering improved contextual understanding and explainability capabilities.

8 Future Work

Although the project achieved strong performance in sentiment classification, several improvements and extensions can be explored in future work. One possible extension is experimenting with larger and more advanced transformer architectures such as BERT, DeBERTa, or GPT-based models to further improve contextual understanding and classification accuracy. Extensive hyperparameter tuning, including optimization of learning rate, batch size, sequence length, and number of training epochs, could also be performed to enhance model performance.

Another important direction is multilingual sentiment analysis, where transformer models can be trained to classify reviews written in different languages. The project can also be extended to aspect-based sentiment analysis, which identifies sentiment toward specific aspects of a review rather than predicting overall sentiment only. For example, a movie review may contain separate opinions about acting, storyline, or cinematography.

In addition, future work may include implementing more advanced explainability techniques to better understand transformer model decisions and attention patterns. Real-time deployment through a web application or API could also make the system more practical for real-world use cases such as customer feedback analysis and social media monitoring. Finally, experimenting with larger datasets and more diverse domains could improve the generalization ability of the models.

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